Peter Dietrich
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10 CFR 50.73

October 18, 2023
NRC-23-0069

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2023-003-00

Pursuant to Title 10 Code of Federal Regulations (10 CFR) 50.73(a)(2)(ii)(A) and 10 CFR 50.73(a)(2)(i)(B), DTE Electric Company (DTE) is submitting LER No. 2023-003-00, "Reactor Pressure Boundary Leakage at Socket Weld on Reactor Recirculation Sample Line Caused by Fatigue Failure".

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2023-003-00, "Reactor Pressure Boundary Leakage at Socket Weld on Reactor Recirculation Sample Line Caused by Fatigue Failure"

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

Enclosure to
NRC-23-0069

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2023-003-00
"Reactor Pressure Boundary Leakage at Socket Weld on Reactor Recirculation Sample Line Caused by Fatigue Failure"

NRC FORM 366 (10-01-2023) | U.S. NUCLEAR REGULATORY COMMISSION | APPROVED BY OMB: NO. 3150-0104 | EXPIRES: 03/31/2024

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Fermi 2	☒ 050 ☐ 052	00341	1 OF 3

4. Title
Reactor Pressure Boundary Leakage at Socket Weld on Reactor Recirculation Sample Line Caused by Fatigue Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
08	20	2023	2023	- 003 -	00	10	19	2023	N/A	☐ 050
									Facility Name	☐ 052 Docket Number

9. Operating Mode	10. Power Level
3	0

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☑ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Eric Frank - Manager, Nuclear Licensing	734-586-4772

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	AD	PSP	X999	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 8/20/2023 at 1600 Eastern Daylight Time (EDT), a Reactor Coolant System (RCS) pressure boundary leak (approximately 2 gpm) was identified at the socket weld on a Reactor Recirculation Sample Line. The small-bore piping experienced vibration stresses at the header location which contributed to the fatigue failure at the socket weld connection. The leak was repaired, and the unit returned to power operation on 9/9/2023. The corrective action plan addressed the causal factors associated with this event. Corrective actions completed or planned include 1) repairing the leak, 2) evaluated if the repair would perform its design function until the next refueling outage in 2024 (RF22), and 3) design change to add adequate supports to alleviate vibration to be installed in RF22. This event had no effect on the public health and safety. The event was reported to the Nuclear Regulatory Commission on 8/20/2023, via Event Notification 56685.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

INITIAL PLANT CONDITIONS

MODE – 3 (Hot shutdown)
Reactor Power – 0

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On 8/20/2023 at 1600 Eastern Daylight Time (EDT) during plant walkdowns in the Drywell while in MODE 3 to identify a cause of increasing unidentified leakage rate, a Reactor Coolant System (RCS) pressure boundary leak (approximately 2 gallons per minute (gpm)) was identified at the socket weld on the 3/4-inch Reactor Recirculation (RR)[AD] Sample Line [PSP] (6DI-B31-7255-1) at the RR header, CT-B31L018. The RR system consists of the two pump loops external to the Reactor Pressure Vessel (RPV)[RPV]. These loops provide the piping path for the driving flow of water to the RPV jet pumps (located in the RPV). The RR system loops are part of the nuclear system process barrier and are located inside the primary containment structure. The recirculated coolant consists of saturated water from the steam separators and dryers that has been subcooled by incoming feedwater. This water passes down the annulus between the RPV wall and the core shroud. A portion of the coolant flows from the RPV, through the two external RR loops and becomes the driving flow for the jet pumps. Each of the two external RR loops discharges high pressure flow into an external manifold from which individual recirculation inlet lines are routed to the jet pump risers within the RPV. The process sampling test line 6DI-B31-7255-1 for chemical tap runs between the manifold header connection and the RPV penetration X-29A. Restraints for the recirculation system piping are the GE cable-type restraints. The steam and recirculation piping are designed in accordance with the 1969 American National Standards Institute (ANSI) B31.7 Nuclear Power Piping Code, Class 1, including addenda ANSI B31.7b-1971.

The RR System circulates the required coolant through the reactor core to maintain fuel temperature within the specified design margins for normal and abnormal operating conditions, and transients. The RR system has the following Safety Functions: 1. Ensure adequate margins for fuel temperature during postulated transients, 2. Maintain pressure integrity during adverse combinations of loadings and forces which occur during abnormal, accident, and special event conditions, and 3. Maintain the ability of the reactor pressure vessel (RPV) to provide a floodable volume after a pipe break accident. The RR system meets the following power generation design bases: a. RR system shall provide sufficient flow to remove heat from the fuel, and b. System design shall minimize maintenance situations that would require core disassembly and fuel removal.

The extent of leaking will result in loss of design function of the pipe, pipe supports, and the equipment attached to the piping. In addition to this, the unit enters a Limiting Condition for Operation (LCO) 3.4.4, which states RCS operational LEAKAGE shall be limited to (a) No pressure boundary LEAKAGE (b) </= 5 gpm unidentified leakage (c) </= 25 gpm total LEAKAGE averaged over the previous 24-hour period; and (d) </= 2 gpm increase in unidentified LEAKAGE within the previous 24-hour period in MODE 1. Technical Specification (TS) 3.4.4 is applicable in MODES 1, 2, and 3. Since the leakage location was identified, this required entry into LCO 3.4.4 "RCS Operational LEAKAGE" CONDITION C – Identification of pressure boundary leakage with a REQUIRED ACTION to be in MODE 3 in 12 hours and MODE 4 in 36 hours. At 1630, a TS-required shutdown to MODE 4 – Cold Shutdown was initiated. The plant was in MODE 4 at 0310 on 8/21/2023. The plant was placed in MODE 5 on 8/30/2023 to support the repair of the leak and the unit returned to power operation on 9/9/2023.

Although the source of the RCS LEAKAGE was identified on 8/20/2023, there was indication that the leak began around 6/10/2022 when RCS UNIDENTIFIED LEAKAGE increased and was recorded in the corrective action program under CARD 22–27022. From 6/10/2022 until 8/20/2023, the UNIDENTIFIED LEAKAGE was within the bounds permitted by

NARRATIVE

LCO 3.4.4 b.-d.; therefore, the plant met LCO 3.4.4. On 8/18/2023, the plant started down-powering to investigate the source of the unidentified leakage. A past operability review determined that this condition is likely to have existed since 6/10/2022, thus the unit was not shutdown within a time frame commensurate with the applicable TS 3.4.4.a requirement of no pressure boundary LEAKAGE. If a design basis accident were to have occurred during this period, the contribution of the 3/4-inch sample line from the CT-B31L018 header would not have contributed significantly to the accident scenario and would have been mitigated in the plant's response to small or large break Loss of Coolant Accidents (LOCAs).

CAUSE OF THE EVENT

The small-bore piping experienced vibration stresses at the header location, CT-B31L018, which contributed to the fatigue failure of the socket weld connection.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

Event notification (EN 56685) was made to the NRC reporting a 4-hour notification per Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(2)(i) (The initiation of any nuclear plant shutdown required by the plant's TS) and 10 CFR 50.72(b)(2)(xi) (A news release or notification of other government agencies) and an 8-hour event notification was made based on meeting the reporting criteria of 10 CFR 50.72(b)(3)(ii)(A) as a condition resulting in the pressure boundary being seriously degraded. Since the plant was already in hot shutdown (MODE 3), the 4-hour notification per 10 CFR 50.72(b)(2)(i) was unnecessary per NUREG 1022 Section 3.2.1, which states "The 'initiation of any nuclear plant shutdown' does not include mode changes required by TS if they are initiated after the plant is already in a shutdown condition." This Licensee Event Report (LER) is being made under the corresponding requirement in 10 CFR 50.73(a)(2)(ii)(A), as a condition of the nuclear power plant, including its principal safety barriers, being seriously degraded.

Although the source of the RCS leakage was identified on 8/20/2023, there was indication that the leak began around 6/10/2022 when RCS UNIDENTIFIED LEAKAGE increased and was recorded in the corrective action program under CARD 22-27022. From 6/10/2022 until 8/20/2023, the UNIDENTIFIED LEAKAGE was within the bounds permitted by LCO 3.4.4 b.-d.; therefore, the plant met LCO 3.4.4. Once the source of the leak was identified on 8/20/2023, LCO 3.4.4 CONDITION C was entered. A past operability review determined that this condition is likely to have existed since 6/10/2022. The pressure boundary leakage likely existed from this time until 8/20/2023 when the unit was placed in MODE 3. The unit was not shutdown within a time frame commensurate with TS 3.4.4.a. Therefore, this event is also reportable pursuant to 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's Technical Specifications.

This event did not result in any actual nuclear safety consequences. The safety significance of this event was low, no safety margins were challenged or reduced.

CORRECTIVE ACTIONS

The pipe was repaired to design specification. A Technical Evaluation concluded that once the pipe was restored to its design configuration, the piping and its components will continue to operate and perform their design functions until the next refuel outage RF22 (currently scheduled for March 2024).

A subsequent re-analysis of the piping and change package will be done to add adequate supports to alleviate the vibration during the next refuel outage, RF22. This is being tracked in a condition report.

PREVIOUS OCCURRENCES

There have been no LERs or Corrective Action Reports/Condition Reports in the last 23-years related to pressure boundary LEAKAGE.